

BASINMARK: WATERMARKING DIFFUSION LANGUAGE MODELS VIA KEYED RE-DENOISING CONTRAST

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ABSTRACT

Working draft. This document exists to typeset and check the baseline comparison while the experiments are still running; the prose is deliberately minimal. Every number is measured on one TITAN RTX unless a row says otherwise, and rows that have not been measured are marked **pending** rather than left blank or filled in from a paper.

1 BASELINE COMPARISON

Table 1 reports watermark detectability against the text quality actually paid for. The axis that matters is not the largest achievable z : an earlier revision of this work tuned an operating point to $z \approx 5$ and only afterwards measured that it cost $4.8\times$ perplexity. Detection is therefore reported as the true positive rate at a controlled false positive rate, and quality as perplexity relative to a non-watermarked control produced under the same decoding regime.

Two properties of the comparison are worth stating before the numbers. *First, each block of the table has its own non-watermarked control.* The three methods require different decoding regimes — dgMARK uses block decoding with top- k sampling, KGW requires strict left-to-right decoding because its green list is seeded by the preceding token, and BasinMark operates post-hoc on drafts sampled at temperature 0.8 over the full vocabulary — so a single shared baseline would charge the cost of a decoding regime to whichever watermark happened to use it. *Second, detection cost is a column, not a footnote.* dgMARK and KGW detect by hashing token identities and need zero model forward passes; BasinMark’s detector must run the diffusion model itself. Collapsing the comparison to detectability alone would hide the method’s main structural disadvantage.

Reading the table. The strongest baseline is KGW at $\delta = 1$: 0.93 TPR at 1% FPR for $1.03\times$ perplexity against its own control, or $1.20\times$ if the cost of the left-to-right regime it requires is charged to it. dgMARK with 3-beam lookahead reaches 0.86 at $1.23\times$ without needing that regime. BasinMark’s best configuration inside a comparable quality budget reaches 0.10, and its best configuration at any quality reaches 0.50 while paying $2.49\times$. The gap is close to an order of magnitude in detectability at matched quality, which is far too large to attribute to hyperparameter search.

The measured cause is capacity rather than tuning. Across all 36 BasinMark configurations the median carrier position admits exactly *one* token within the fluency budget, and the candidate slate never truncates, so a watermark defined on *which tokens are present* has almost no freedom to spend. This also explains why moving the watermark to generation time made things worse rather than better: the schedule it forces — all non-carrier positions, then all carrier positions — costs $1.98\times$ perplexity on its own, before any watermark is applied.

Method	Setting	PPL ↓	Ratio	TPR at FPR			z	Det. fwd. ↓
				5%	1%	0.1%		
Generation-time watermarks (the watermark is applied while the text is produced)								
No watermark	block dec., top- k 3	9.94	1.00×	—	—	—	+0.04	—
dgMARK (Hong & No, 2026)	$k = 1$	15.40	1.55×	0.88	0.74	0.62	+4.35	0
dgMARK (Hong & No, 2026)	+3-beam	12.23	1.23×	0.92	0.86	0.82	+5.81	0
No watermark	left-to-right	11.61	1.00×	—	—	—	+0.24	—
KGW (Kirchenbauer et al., 2023)	$\delta = 1$	11.96	1.03×	0.97	0.93	0.73	+3.85	0
KGW (Kirchenbauer et al., 2023)	$\delta = 2$	16.21	1.40×	1.00	1.00	1.00	+6.93	0
KGW (Kirchenbauer et al., 2023)	$\delta = 3$	14.05	1.21×	1.00	1.00	1.00	+8.92	0
Post-hoc watermarks (the watermark is applied to text that already exists)								
No watermark	$T = 0.8$, full vocab	26.21	1.00×	—	—	—	−0.03	—
BasinMark	keyed carrier	27.7	1.11×	0.00	0.00	0.00	+0.73	9
BasinMark	entropy gate	41.7	1.59×	0.30	0.20	0.10	+1.49	9
BasinMark	leverage, quality-first	30.5	1.17×	0.10	0.10	0.00	+0.58	9
BasinMark	leverage, detection-first	65.3	2.49×	0.62	0.50	—	+2.94	9
BasinMark	generation-time, two-phase	203.7	7.77×	0.12	0.00	0.00	+1.14	9

Table 1: Watermark detectability against the text quality actually paid for. All rows measured on LLaDA-8B-Instruct, C4 prompts (300-character truncation), 256 generated tokens, GPT-2-large perplexity; $n = 50$ for dgMARK, $n = 30$ for KGW, $n = 10$ for BasinMark. *Ratio* is perplexity relative to the matching non-watermarked arm directly above it. Each block is compared against its own control because the decoding regime differs: dgMARK uses block decoding with top-k 3, KGW requires strict left-to-right decoding (its green list is seeded by the preceding token, which order-agnostic decoding does not provide), and BasinMark operates on drafts sampled at $T = 0.8$ over the full vocabulary. *Det. fwd.* is the number of model forward passes a detection requires. Non-watermarked arms sit at $|z| \leq 0.25$, so the thresholds are not inflated. KGW is scored on each *distinct* bigram: its green list is seeded by the preceding token, and under forced left-to-right decoding LLaDA repeats roughly a third of its bigrams, so re-scoring repetitions gave $z = +3.32$ and a 37% false positive rate on the non-watermarked $\delta = 0$ arm. Two caveats on KGW’s apparently free detectability. Its regime carries a cost the ratio column hides: left-to-right decoding scores 11.61 against 9.94 for block decoding, so measured against the better control $\delta = 1$ costs $1.20\times$, not $1.03\times$. And perplexity is not monotone in δ here (16.21 at $\delta = 2$ versus 14.05 at $\delta = 3$), because a strong green-list bias also makes the text more repetitive, which an n -gram-blind perplexity rewards.

Nominal α	Mean FPR	Worst-key FPR	Verdict
0.10	0.0121	0.0583	valid ($\leq \alpha$)
0.05	0.0037	0.0250	valid ($\leq \alpha$)
0.01	0.0000	0.0000	valid ($\leq \alpha$)

Table 2: False positive control for a fixed key, measured per key. 20 keys $\times$ 120 non-watermarked LLaDA generations = 2400 draws; z over all of them has mean -0.076 and standard deviation 0.976 . The decision rule is the one detection actually uses, $p_{\text{bound}} = \exp(-z^2/2)$, Hoeffding’s inequality for Rademacher sums. Deployment uses one secret key over many documents, so the relevant guarantee is $P_Y(p \leq \alpha \mid K = k) \leq \alpha$ for the *worst* k , not an average over keys, and a randomization p -value owes super-uniformity rather than strict uniformity. An earlier revision reported a Gaussian tail instead and measured 0.145 at $\alpha = 0.10$ — anti-conservative, and the reason the exactness claim was withdrawn.

2 DETECTOR VALIDITY

The detector’s p -value carries no calibration corpus and no model-specific null estimate: exchangeability of the two arms of each challenge makes every per-probe contrast independently symmetric about zero, so conditional on the magnitudes the null is a Rademacher mixture. Exact enumeration is affordable only up to 20 blocks, and every configuration here has 48 or more, so the reported value is Hoeffding’s bound $\exp(-z^2/2)$ — a finite-sample upper bound with no distributional assumption and no resolution floor. A Monte-Carlo randomization estimate is also computed, but it cannot resolve below $1/(1+n)$ and is never used for tail claims.

Table 2 checks the guarantee in the form deployment needs. Each document receives its own key $K_d = \text{HMAC}(K, \text{nonce}_d)$, and false positive rates are reported per key and at the worst key rather than averaged over keys.

What is settled and what is not. The statistical side of BasinMark holds: the null is valid at the worst key, detection cost is L forward passes independent of payload size, and the challenge-response signal is present. The embedding channel is what fails. A watermark defined on *which tokens are present* can only be driven by changing tokens, and on this model the median carrier position admits one token inside the fluency budget, so the channel is priced out. Whether steering commit *order* instead — never changing a candidate token, the property that makes dgMARK nearly free — can carry the same statistic is the open question, and the block-local formulation testing it is running.

REFERENCES

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